

# Who Benefits from Scotland's Alcohol Minimum Unit Pricing?

## A Novel CITS-MAIHDA Framework for Intersectional Inequality Evaluation



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### THE PROBLEM

#### BACKGROUND / CONTEXT

Scotland's Minimum Unit Pricing (MUP) targets heavy drinkers by raising the national floor price of alcohol. Overall consumption reduced, but the **inequalities of benefit** across **population subgroups** remains poorly understood.

#### THE GAP IN EVIDENCE

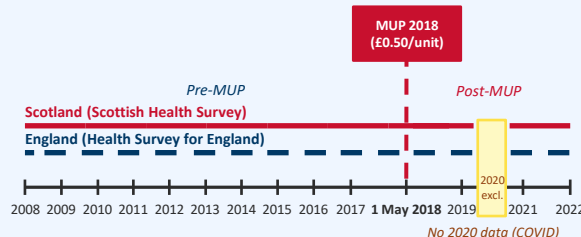
Current MUP evaluations show contradictory results as they examine **age, sex, or deprivation** in isolation. No study has holistically mapped how these **dimensions intersect** to shape both **additive and interaction-driven** inequalities.

#### RESEARCH AIMS

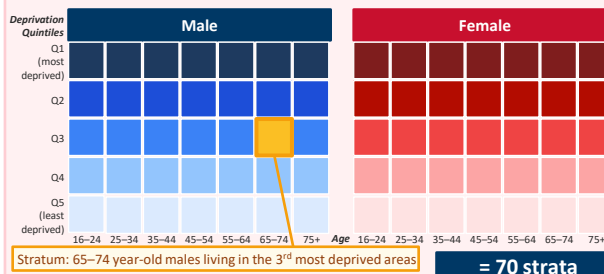
- ① Analyse historical trend in alcohol consumption
- ② Evaluate overall MUP impact on consumption.
- ③ Examine unequal MUP impacts via both isolated and intersectional lens (age × sex × deprivation)

### THE FRAMEWORK

#### CITS DESIGN (Scotland vs England, 2008–2022)



#### MAIHDA: 70 Intersectional Strata (2 Sex × 7 Age × 5 Deprivation)



### STRENGTH & DATA

#### METHODOLOGICAL STRENGTH

- Integrate intersectional MAIHDA (multilevel analysis of individual heterogeneity and discriminatory accuracy) model into CITS design for causal inference
- Derive real-world interaction effect from contrasting Bayesian posterior distributions

#### DATA & DESIGN

- 👤 National Cohorts: Scottish population vs English population
- 📅 Timeline: Annual repeated cross-sectional surveys from 2008 to 2022
- ⚙️ Modelling: Bayesian multilevel MCMC estimation via "brms" (R package)

Chapter 1 (Current Stage): Alcohol consumption

Chapter 2: Alcohol-specific hospitalisations

Chapter 3: Alcohol-specific mortality

### TAKE-AWAY MESSAGE:

MUP reduced drinking overall — but did it reach those who need it most? We are building a novel framework to unmask who benefits, and who is left behind.

Downloadable version & references



# References

## BACKGROUND / CONTEXT

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## METHODOLOGICAL STRENGTH

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## DATA & DESIGN

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